
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Generative image models trained on large-scale datasets now synthesize visually compelling images, yet they continue to fail on concepts that appear elementary and plausibly well represented in their training data, such as human hands or objects appearing in groups of four. Because such failures are often documented through isolated examples, it remains difficult to determine whether they are incidental sampling artifacts or symptoms of a systematic mismatch between the concepts present in natural data and those realized by trained generators. We address this gap by introducing a framework for identifying and characterizing "conceptual blindspots": concepts that occur in the data distribution but are substantially suppressed or exaggerated in a model's generations. The method uses sparse autoencoders (SAEs) to derive interpretable concept embeddings, enabling quantitative comparisons of concept prevalence between natural and generated images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—which supports fine-grained measurement of conceptual disparities at both distributional and datapoint levels. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the framework surfaces concrete suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual datapoint level, the same representation isolates memorization artifacts—cases in which models reproduce highly specific visual templates that appear in the natural reference set. Together, these results establish a theoretically grounded and empirically operational framework for assessing the conceptual fidelity of generative image models relative to a reference data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have reached a level of visual competence that now supports applications within computer vision [1–9] and across adjacent domains [9–15]. This progress nevertheless coexists with persistent failures on concepts that appear visually and linguistically simple, including human hands [16–19], objects appearing in groups of four [20], and negations or object relations [21–22]; these limitations are documented by both qualitative [23–25] and quantitative studies [22, 26]. When prompted with such concepts, models often generate adjacent visual structures rather than the intended concept itself—for example, producing hands with six fingers rather than anatomically plausible hands. We refer to these failures as "conceptual blindspots"¹—cases in which exposure to relevant visual evidence in the training distribution does not translate into faithful generation. The central question is therefore not whether individual

¹We borrow the term "blindspots" from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision's rationale.



Figure 1: $\delta(k)$ quantifies whether a model under- or over-generates concept c_k relative to its prevalence in natural data. We treat concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. Images from four popular generative image models illustrate both conceptual blindspots and concepts whose prevalence remains aligned with the natural baseline.

prompts expose striking defects, but whether such failures instantiate a broader phenomenon in which trained generators systematically underproduce or overproduce particular concepts relative to a natural reference distribution.

Existing evaluation protocols for generative image models do not directly answer this concept-level question [28, 29]. Many standard metrics, including FID [30], primarily summarize image realism or global distributional proximity, thereby obscuring failures localized to particular semantic or visual concepts. CLIPScore-style evaluations and diversity or coverage statistics [31–33] provide complementary evidence, but they typically operate above the granularity at which fine-grained features and concepts can be isolated [34, 35]. Human studies and open-ended qualitative audits [5, 24, 36–39] can expose salient failures, yet they do not by themselves supply a scalable, repeatable basis for comparing conceptual misalignment across model families.

This work. We develop an automated, unsupervised methodology for eliciting concepts whose prevalence differs between samples from a natural reference process and samples produced by a trained generative image model. The paper makes three connected contributions:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We define conceptual blindspots as concept-level deviations between a generative model and a natural data-generating process, and we introduce a quantitative criterion for identifying suppressed and exaggerated concepts (Section 3). This formulation shifts the analysis from anecdotal failures or manually selected concepts to a principled distributional comparison.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We operationalize the formal definition with sparse autoencoders (SAEs), which extract interpretable concept representations from natural and synthesized images and support direct comparison of their concept distributions (Section 4). The analysis is enabled by an archetypal SAE (RA-SAE) trained on DINOv2 features with 32,000 concepts, the largest such RA-SAE to date, which we open-source.
- **Exploratory Tool and Empirical Analysis.** We provide an exploratory web tool for distribution- and datapoint-level inspection of blindspots across models (Sections 5.1–5.4). Applying the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5) reveals specific suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), as illustrated in Fig. 1.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early work in explainable AI for computer vision largely attributed model behavior to influential input regions [40–42]. Although useful for localization, such methods provide limited semantic access to learned representations and can yield unreliable explanations [43–45]. Concept-based interpretability [46] addresses this limitation by identifying semantically meaningful directions in neural networks, thereby characterizing not only where a model attends but also which concepts and structures its representations encode [47–49].

A growing body of work shows that several prominent concept-based methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be understood as variants of dictionary learning with different constraints [52]. Within this family, SAEs provide a particularly scalable mechanism for concept-based interpretability. Although standard SAEs can produce overly specific features [53], encounter compositionality challenges [54], and support only limited intervention effects [55], recent variants improve their suitability for structured concept analysis; in this work we use archetypal SAE (RA-SAE) [56].

Other interpretability and evaluation approaches, including prompt-based probing [57], attention-map or activation visualizations [58, 59], and dataset-level statistics [32, 33], provide only partial evidence for concept-level failures. Their unit of analysis is typically an image, prompt, or aggregate distributional score rather than a fine-grained feature or concept [34, 35]. Moreover, many such analyses require subjective interpretation and do not separate distinct failure modes [60]. This leaves an important methodological gap: existing metrics do not systematically identify which concepts are suppressed or exaggerated by generative image models [28].

2.2 Generative Image Models

Diffusion-based methods now constitute a central modeling paradigm across generative vision, including image [4, 61–64], video [65–72], and 3D generation [7, 73–75]. For images, this trajectory begins with denoising diffusion probabilistic models (DDPMs) [76] and extends through non-Markovian denoising diffusion implicit models (DDIMs) [77].

Stable Diffusion (SD) [9] made latent diffusion models broadly accessible to both research and open-source communities. Subsequent releases, including SD 2 [78], SD 3 [79], and SD XL [80], were accompanied by architectural extensions and conditioning mechanisms for style [81, 82], pose [83], identity [84, 85], and additional input modalities such as image-to-image generation. Related latent diffusion systems, including Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89], further broaden the model families for which concept-level evaluation is relevant.

2.3 Datasets for Generative Image Models

The performance of contemporary generative vision models is inseparable from large-scale computation and internet-scale image-text corpora [90–92]. LAION-5B [93], scraped from Common Crawl [94], has been especially influential for open-source text-to-image systems, including SD and its derivatives; it contains more than 5B image-caption pairs, 2.3B of which are in English. Other major resources include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M image-caption pairs, respectively.

As LAION-5B gained popularity, concerns grew over its biases [97–101]. Despite filtering attempts, harmful content persisted [97–99], including NSFW material [100] and hundreds of CSAM instances [101], prompting its temporary removal from official channels. The dataset also suffers from low-quality images [102] and internet-style captions (e.g., product descriptions) that misalign with how users prompt trained models [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning constructs sparse representations in which each input is reconstructed from a small number of learned atoms [104, 106]. Building on compressed sensing [107, 108], the field has progressed from vector quantization [109] to methods such as Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. More recent work develops online learning [114], structured sparsity [115], theoretical guarantees [116, 117], and connections to deep learning [118, 119].

Sparse coding has also been used to study the emergence of high-level concepts inside diffusion models [120]. Before diffusion models, concept-grounded interpretability for generative systems relied on concept-bottleneck models that require human intervention during training [121] or post-hoc detectors that retrofit concept supervision [122]. Because both approaches depend on human-defined concept sets, they can miss systematic trends that are not specified in advance.

Aggregate metrics such as precision, recall, density, and coverage [35, 123], as well as latent density scores that estimate sample quality from a model’s latent space [124], are valuable for evaluating

generative image models. However, even when these metrics reveal distributional gaps, they do not identify the specific concepts that are under- or over-represented.

These limitations motivate a scalable, unsupervised framework that identifies and quantifies concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We formalize *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and the conceptual content realized by model-generated outputs. This formalization yields quantitative criteria for determining which concepts are under- or over-represented by a generative model relative to the data distribution. The construction is summarized in Fig. 2. Although the derivation uses standard assumptions [126–130], the empirical analyses below show that the resulting measurements elicit meaningful phenomenology even when those assumptions hold only approximately.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(\mathbf{c}) = \exp(-E(\mathbf{c}))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(\mathbf{c}) = \sum_k E(c_k)$ and hence $p(\mathbf{c}) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $\mathbf{c} \in \mathcal{C}$ to observations $\mathbf{x} \in \mathcal{X}$, i.e., $\mathbf{x} = G(\mathbf{c})$.

For notational economy, $p(\cdot)$ denotes both the latent density $p(\mathbf{c})$ and its pushforward to image space $p(\mathbf{x})$, with $\mathbf{x} = G(\mathbf{c})$. The invertibility of G justifies this convention because it induces a valid distribution over $\mathcal{X}$ through the change-of-variables formula. The latent coordinates therefore represent the **Concepts** that structure the data distribution $P_{\mathcal{X}}$ over the image observation space $\mathcal{X}$. Such coordinates may correspond, for example, to color, shape, size, location, and related visual factors [131–132]. We allow the data-generating process to associate each sampled image with a text description $\mathbf{t} \in \mathcal{T}$, while leaving the captioning mechanism implicit. A text-conditioned **Generative image model** g_{θ} , with parameters θ , is then trained on image-text pairs to map a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a scene description $\mathbf{t}$ to an image $\mathbf{x}$ intended to depict that description.

To identify conceptual blindspots in g_{θ} , we compare the probability mass assigned to each concept by the data-generating process with the corresponding mass induced by the model’s generations. We therefore take an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and construct a synthetic dataset $\mathcal{D}'_{\mathcal{X}}$ by generating images from the associated text descriptions. Concept occurrence is estimated with an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to a d -dimensional vector whose k^{th} coordinate is the scalar energy of concept k , i.e., $\xi_k(\mathbf{x}) = E(c_k)$. The invertibility assumption on the data-generating process makes this concept-energy estimate well defined. Accordingly, the probability associated with image $\mathbf{x}$ can be written as $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (with partition function Z_{ξ}), yielding the dataset-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{\mathbf{x} \in \mathcal{D}_{\mathcal{X}}} p_k(\mathbf{x})$. Using this quantity and the sigmoid map $\sigma(\cdot)$, we define the average energy difference for concept k between $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $\mathbf{x} \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $\mathbf{x}' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the

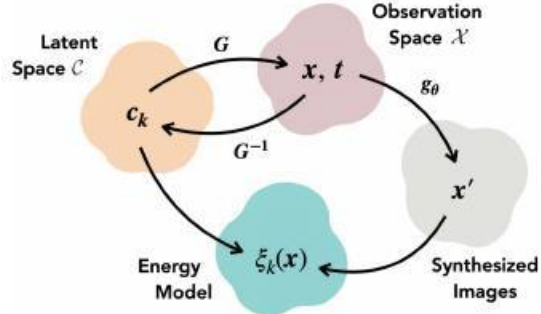


Figure 2: **Framework Mapping.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations $(\mathbf{x}, \mathbf{t})$ by the invertible DGP. S_{θ} generates $\mathbf{x}'$ conditioned on $\mathbf{t}$. The energy model $\xi_k(\mathbf{x})$ extracts comparable concept representations from $\mathbf{x}$ and $\mathbf{x}'$, making concept-distribution discrepancies observable as blindspots.

178 energy difference for concept k as:

$$\begin{aligned}\delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})},\end{aligned}\tag{1}$$

179 where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
180 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

181 The energy difference for concept k therefore measures the relative probability of that concept in
182 the generated observations $\mathcal{D}'_{\mathcal{X}}$ against the natural baseline $\mathcal{D}_{\mathcal{X}}$. This measure directly induces the
183 following definition of conceptual blindspots.

184 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
185 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
186 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

187 A conceptual blindspot is thus a concept whose generated prevalence departs markedly from its
188 prevalence under the data-generating process, either through suppression or exaggeration. Suppressed
189 concepts have disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts
190 are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). In all experiments, we use $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to
191 isolate these two regimes.

192 This definition is related to “mode collapse” as studied in prior work (e.g., Bau et al. [39]), but
193 differs in its granularity. Mode-collapse analyses examine exaggerated or suppressed probabilities
194 of generating *entire images*; our definition instead measures changed probabilities for individual
195 concepts. For example, when a model fails to generate an object with a `white` background, the
196 relevant concept is a suppressed conceptual blindspot.

197 4 Method: Operationalizing the Definition of Conceptual Blindspots

198 We now instantiate the preceding definition as a pipeline for identifying conceptual blindspots in a
199 generative model g_θ . Following Sec. 3, the pipeline requires (i) model-generated images that can
200 be paired with samples from the data-generating process and (ii) an energy model that maps both
201 image sets into a shared concept basis. Below, we use $\|\cdot\|_F$ to denote the Frobenius norm and $\|\cdot\|_0$
202 to denote the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$
203 implies element-wise non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$, and denote the i -th row of
204 a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

205 **From Prompts to Latent Representations.** For a model g_θ , we compare natural image-caption
206 pairs $(\mathbf{x}, t)$ from $\mathcal{D}_{\mathcal{X}}$ with synthetic counterparts generated by conditioning g_θ on the same text
207 descriptions.

208 Given a caption t , we synthesize $\mathbf{x}'$ with a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$
209 implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling occurs in
210 latent space via a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

211 where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0
212 is decoded via a pretrained VAE to yield the synthetic image $\mathbf{x}' = \text{VAE}(\gamma_0)$. For the remainder of
213 the paper, we treat g_θ as a black box that maps prompts to images: $t \mapsto \mathbf{x}'$.

214 **Defining the Energy Model.** Prior work shows that self-supervised learning methods can invert
215 aspects of the data-generating process and identify the underlying energy function up to linear
216 transformations [126, 128, 133, 134]. We therefore use DINOv2 [135], a state-of-the-art self-
217 supervised vision model, as the feature backbone for our energy model. Because the number of
218 concepts underlying the DGP may exceed the dimensionality of the model feature space [51, 136],
219 we train sparse autoencoders (SAEs) on DINOv2 features to recover subspaces associated with
220 those concepts [50, 56, 137, 139]. If DINOv2 represents generative concepts along approximately
221 orthogonal directions, consistent with the independence constraint in Def. 1, then an SAE can isolate

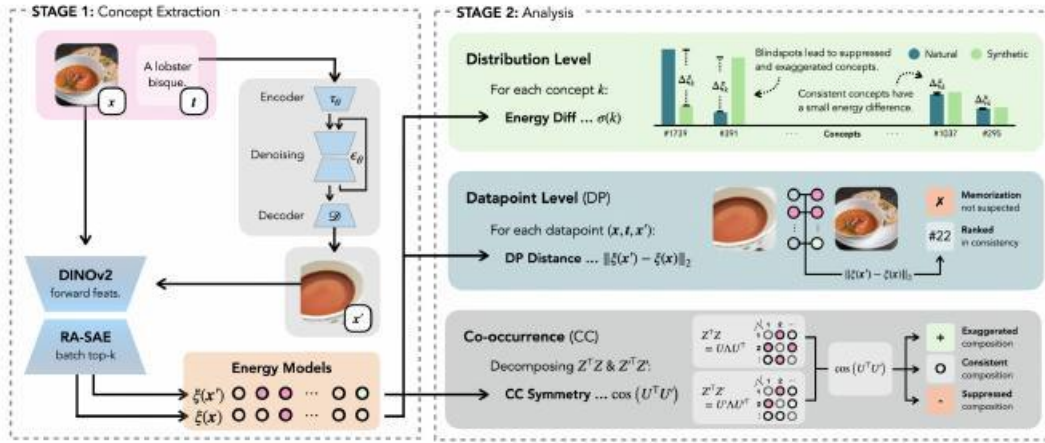


Figure 3: **Concept Extraction Pipeline.** For each triplet (x, t, x') , DINOv2 features are extracted from the real and generated images and processed by real concept embeddings, yielding $\xi(x)$ and $\xi(x')$. The coordinate $\xi_i(x) = \xi_i(x')$ represents the energy of the i -th concept. These concepts are then mapped into individual latent dimensions (Fig. 3). The activation of an SAE dimension then serves as our approximation to the ground-truth energy assigned to the corresponding concept.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor. For natural and synthetic images from $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$, we compute features $a = f(x) \in \mathbb{R}^d$. For n image-text pairs, the real and generated features are stacked into matrices $A, A' \in \mathbb{R}^{n \times d}$. An SAE then decomposes each feature vector into a sparse combination of learned concept atoms. Let $D \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the encoder that maps features to sparse codes. Applying Ψ row-wise yields $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, where each row $z_i = \Psi(a_i)$ is the concept decomposition of an image. The decoder reconstructs features as ZD^T , and training minimizes reconstruction error under sparsity and non-negativity constraints:

$$\min_{\Psi, D} \|A - \Psi(A)D^T\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs can converge to arbitrarily oriented dictionaries, which makes downstream concept analyses sensitive to random initialization. To stabilize the learned basis and improve reproducibility, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary D to lie in the convex hull of the training data. Specifically, we write $D = WA$ with $W \in \Omega_{K', n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Thus every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and every reconstruction ZD^T lies in the conic hull $\text{cone}(A)$. This constraint keeps learned concepts anchored to the support of the data distribution [56]. After training, the SAE produces a common sparse code system: Z for real images and Z' for their generated counterparts. These codes capture the same prompt-conditioned visual semantics in terms of shared, interpretable concepts, with the activation value of the concept serving as energy values for our analysis of conceptual blindspots. Consequently, for each prompt-image pair (t, x) , the method synthesizes x' and maps both images into a shared sparse concept space, yielding interpretable vectors (z, z') that support direct measurement of conceptual shifts induced by generation.

5 Results

We evaluate four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and matched generations (Fig. 4). The analysis proceeds at three complementary resolutions (Fig. 3): a **distribution-level** evaluation identifies suppressed and exaggerated concepts; a **datapoint-level** analysis distinguishes failures associated with ambiguity, omission, and memorization; and a **compositional** analysis



Figure 4: **Representative Matched Generations.** LAION-5B image-text pairs and matched generations from four models.

ysis characterizes distortions in concept co-occurrence geometry.

A central output of the framework is the interactive exploratory tool in Fig. 5. The web interface organizes the RA-SAE concept space through UMAP projections, allowing users to compare concept-level energy differences across models. It is publicly available at a software repository together with pre-computed energy-difference data for SD 1.5, SD 2.1, PixArt, and Kandinsky. The analyses below are derived from the following capabilities:

- **Contrast models and architectures.** For each evaluated model, the tool displays a UMAP visualization over all 32,000 RA-SAE concepts, with each concept color-coded by its energy difference.
- **Inspect individual concepts.** Each concept card reports key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Explore blindspots.** Global rankings of suppressed and exaggerated concepts complement the UMAP and per-concept views, making the most salient blindspots directly inspectable.

Rather than exhaustively analyzing a single failure mode, we use the tool to demonstrate how the framework supports broad, customizable exploration of conceptual misalignment.

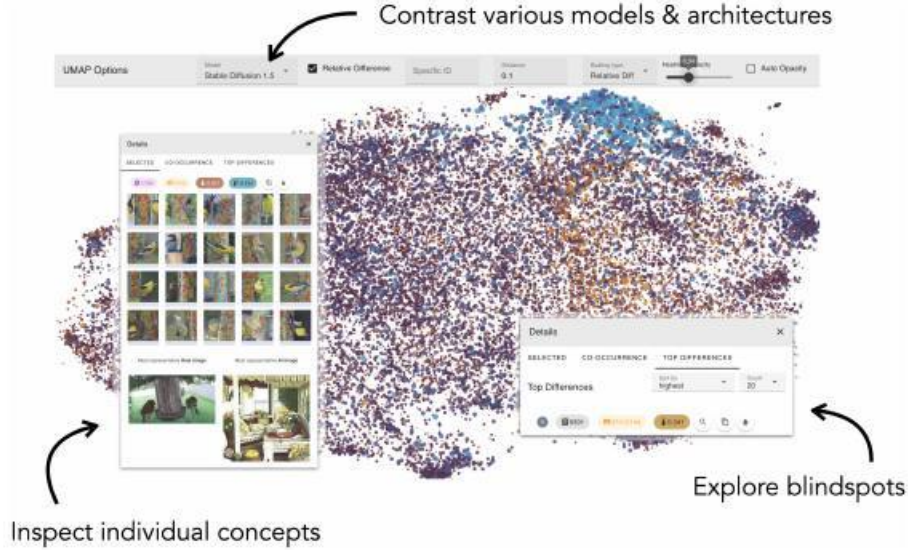


Figure 5: **Overview of the Exploratory Tool.** For each evaluated model, the interface displays a UMAP projection in which each dot is a concept colored by its energy difference. Selecting a concept opens a detail panel with illustrative images, statistics, representative natural and generated images x and x' , ordered co-occurrences, and global blindspot rankings.

5.1 Generative Image Models Cannot Produce Unseen Concepts

We first quantify model-data disparities by measuring the marginal energy difference $\delta(k)$ for each of the 32,000 RA-SAE concepts. As defined in Sec. 4, $\delta(k)$ measures the relative prevalence of concept k in synthesized images compared with natural images: $\delta(k) < 0.1$ marks suppression, whereas $\delta(k) > 0.9$ marks exaggeration. Fig. 6 reports the resulting distributions for all four evaluated models. Every model exhibits a heavy-tailed distribution with mass at both extremes, showing that concept coverage errors are not isolated to a small number of concepts. The left tail, corresponding to suppressed concepts, is consistently denser and longer than the right tail, indicating a systematic tendency toward concept omission. This asymmetry is captured by the negative skewness of the distributions: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

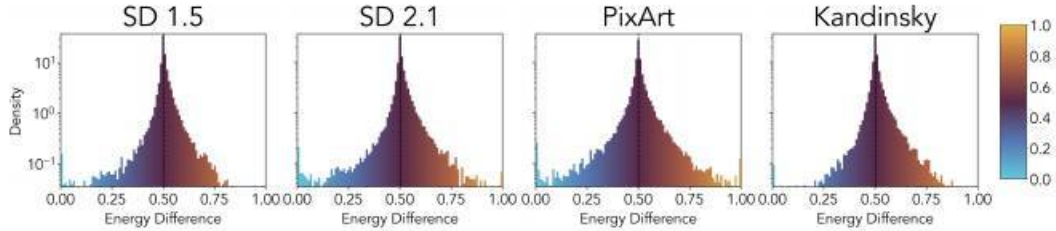


Figure 6: **Concept Energy Distribution.** Log-scale histograms of $\delta(k)$ over 32,000 concepts compare each model’s synthesized distribution with the natural distribution. Values left of zero denote suppressed concepts, whereas values right of zero denote exaggerated concepts.

285 All models therefore contain both suppressed and exaggerated concepts, but the shape of the discrepancy profile varies by model. PixArt, for example, exhibits a wider spread, consistent with a more
 286 dispersed suppressed-concept distribution. The shared left-skew establishes concept omission as a
 287 recurring pattern, while the model-specific profiles motivate the structural analyses that follow.
 288

289 5.2 Structure and Specificity of Conceptual Blindspots

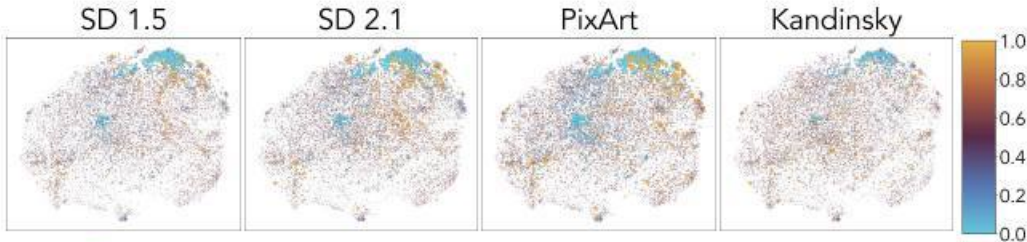


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts are colored by $\delta(k)$ between natural and synthesized distributions. Clustered blue regions indicate suppressed concepts, while yellow regions indicate exaggerated concepts.

290 Having established marginal discrepancies, we next examine whether blindspots are organized in
 291 concept space. We embed all 32,000 concepts into two dimensions with UMAP on the sparse codes
 292 and color each concept by its $\delta(\cdot)$ value. Fig. 7 shows that concept-level discrepancies form distinct
 293 clusters across all evaluated models. These clusters frequently appear as contiguous blindspot regions,
 294 particularly among suppressed (blue) concepts, indicating that blindspots have structured geometry
 295 rather than arising as independent concept-level accidents. We then quantify how consistently these
 296 structures transfer across model families by comparing concept-level $\delta(\cdot)$ vectors. Fig. 8 reports
 297 scatter plots and pairwise Pearson correlations between SD 1.5 and the other evaluated models. SD
 298 1.5 and SD 2.1 are strongly aligned ($r = 0.82$), consistent with their related architectures and training
 299 pipelines. By contrast, correlations with PixArt and Kandinsky are substantially lower— $r = 0.41$
 300 and $r = 0.46$, respectively—showing that different architectures emphasize different regions of the
 301 concept space.

302 Overall, the analysis above reveals that while some blindspots are universally shared—likely due to
 303 properties of the dataset—others are highly model-specific, emerging from idiosyncrasies in training
 304 dynamics or model capacity. This motivates the need to identify and study both blindspots that are
 305 shared across models and ones that are unique to specific models in subsequent sections.

306 5.3 Qualitative Examples of Blindspots

307 We next visualize specific examples of both suppressed and exaggerated blindspots to gauge what
 308 concepts fall under these regimes. Specifically, in Fig. 9a we show an example of a conceptual
 309 blindspot suppressed by all models—we find all evaluated models fail to reproduce the concept `solid`
 310 `white` on documents. As can be seen in the figures, despite the caption explicitly referencing this
 311 concept, none of the generated images reflect the intended visual semantics, suggesting that this
 312 region of the concept space is systematically under-sampled across models. Meanwhile, Fig. 9b
 313 highlights a model-specific blindspot: the concept `pan` is accurately captured by three models, yet

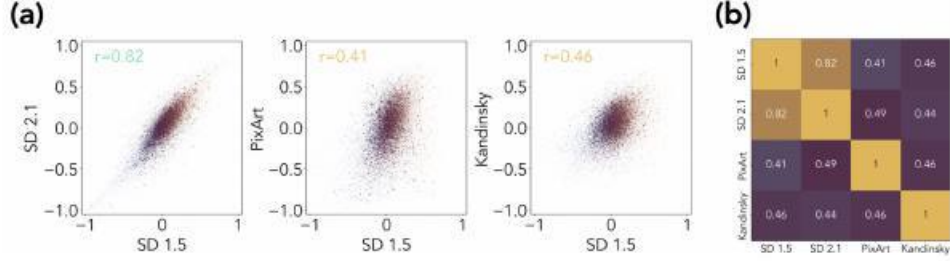


Figure 8: **● Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported top left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations among other architectures, indicating model-specific blindspots. **● Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for δ between all models, quantifying the degree of shared conceptual blindspots across these models.

conspicuously missing from generations produced by Kandinsky. This reinforces the findings from Sec. 5.2 where cross-model agreement was found to be high in some cases but limited in others.

Conversely, in Fig. 10 we present a case of exaggeration, where the concept shadow under animal is overly emphasized in generated images. While shadows are mildly plausible, their consistent and pronounced rendering across models, relative to the more nuanced and variable occurrences in natural images, suggests an overactive prior. Interestingly, despite attempts at finding concepts that are uniquely exaggerated by a specific model, we did not find any clear examples—this suggests exaggerations are approximately universal.

Overall, the examples above concretely demonstrate how conceptual blindspots manifest in generated outputs, illustrating that our energy-based diagnostic can surface both shared and model-specific failure modes. Notably, it enables the identification of surprising model limitations—such as the consistent failure to reproduce clear or solid background elements, like whitespaces on documents, across all models. This raises the possibility that certain failure patterns may stem from architectural constraints or training data biases that transcend individual model idiosyncrasies.

While these aggregate-level analyses are informative, they invite a deeper question: do these blindspots emerge only in the aggregate across many samples, or



Figure 10: **● Example of an Exaggerated Conceptual Blindspot.** Four synthesized images $D'_{\mathcal{X}}$, archetypal for the concept shadow under animal, alongside the corresponding natural im-

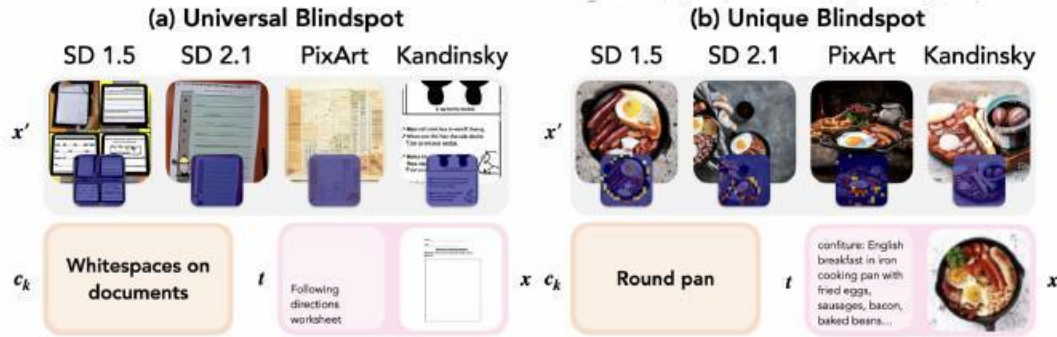


Figure 9: **● Examples of Suppressed Conceptual Blindspots.** The natural images x , representative c_k and t , shown alongside four synthesized images x' , generated using S_θ . The universal blindspot is present in all evaluated models; the unique blindspot is only present in Kandinsky.

do they manifest themselves even at the level of individual datapoints? This finer-grained perspective allows us to probe the mechanisms behind blindspots more directly—uncovering cases of prompt misinterpretation, latent memorization, or both.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine individual natural vs. generated image pairs for which the $\delta(\cdot)$ values averaged across all concepts exhibit the largest and smallest differences. This analysis aids easy understanding of model success and failures, latter of which we find often arises from prompt ambiguity or memorization artifacts. For example, Fig. 11a shows instances with near-zero difference in average $\delta(\cdot)$ values. In these cases, the generated images are conceptually indistinguishable from the original. However, qualitative inspection clearly shows this happen not because the model faithfully captures the prompt semantics, but from pure replication of memorized templates: we see repetitive visual structures (e.g., outlines of clothing or object arrangements), indicating that the model may be copying from overly frequent patterns in the training data. By contrast, Fig. 11b illustrates samples that are among the largest $\delta(\cdot)$ values. These indicate significant conceptual divergence between the synthesized and natural image. While some of these discrepancies can be attributed to underspecified or noisy captions, others reveal genuine blindspots: the prompt describes a clear concept faithfully present in $\mathcal{D}_{\mathcal{X}}$, yet the model fails to realize it in $\mathcal{D}'_{\mathcal{X}}$. This failure suggests that even when language grounding is adequate, certain concepts fall outside the model’s generative abilities.



Figure 11: **Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences where models appear to memorize training patterns. (b) Examples with large differences where significant concept divergences due to prompt ambiguity or model limitations occur.

374 5.5 Analyzing Post-Training Effects

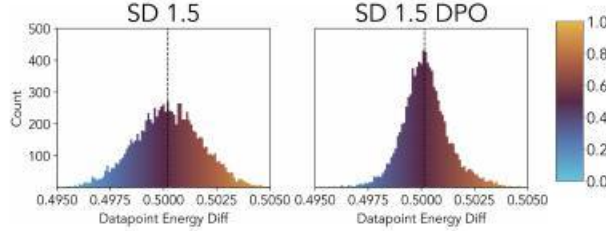


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distribution of SD 1.5 models with and without DPO.

375 Post-training protocols, e.g., safety fine-tuning, have been argued to reduce the diversity of model
 376 generations [140]. Given our pipeline’s ability to isolate interesting differences in a model’s gen-
 377 erations and the ground-truth DGP, we next use it to understand the effects of DPO—a popular
 378 safety fine-tuning protocol [141]. Specifically, we compare two variants of the SD 1.5 model: one
 379 trained with DPO, and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of
 380 the difference between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents
 381 a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits both a
 382 lower median and a narrower spread, indicating more consistent distribution of generated concepts
 383 with the ground-truth DGP. This suggests that DPO may serve to regularize the model’s concept
 384 distribution, encouraging outputs that more closely reflect the semantic content of the seen inputs.
 385 While our analysis does not disentangle the specific inductive biases introduced by DPO, these results
 386 provide empirical evidence that its optimization objective, which favors human-preferred generations,
 387 indirectly promotes better match with the training distribution. In particular, it reduces *both over and*
 388 *under* activation of individual concepts relative to the natural baseline. These findings highlight the
 389 utility of our pipeline in characterizing the downstream effects of post-training interventions: not
 390 merely in terms of output quality, but in how they reshape the conceptual geometry of the model’s
 391 output space.

392 5.6 Conceptual Misalignment as a Function of Empirical Frequency

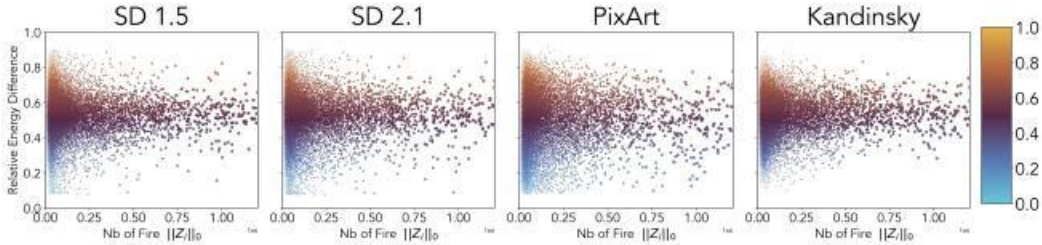


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots showing the relationship between concept frequency (x-axis) and the energy difference (y-axis) across four evaluated models. Each point represents a concept with size is proportional to its activation frequency.

393 We previously hypothesized that conceptual blindspots are not merely be architectural artifacts,
 394 but may also emerge as a direct consequence of distributional peculiarities of certain features. In
 395 this section, we empirically test this hypothesis by examining whether concepts that are rarely
 396 activated in natural images, i.e., those in the long tail of the data distribution, lead to blindspots
 397 in generative models. Specifically, we process the natural dataset $\mathcal{D}_X$ through the trained SAE
 398 and compute, for each concept k , its empirical frequency $\|\mathbf{Z}_{:,k}\|_0$, where $\mathbf{Z}_{:,k}$ is the activations of
 399 concept k across all our images. We then correlate this with the absolute energy difference observed
 400 across generated outputs. Fig. 13 visualizes this relationship for all evaluated models. We find
 401 that concepts with higher frequency in natural data tend to show lower energy discrepancies, while
 402 rare concepts—especially suppressed ones ($\delta(k) < 0.5$)—exhibit significant alignment errors. This
 403 suggests that many blindspots stem not from randomness or model quirks, but from systematic effects
 404 tied to long-tail concept distributions. Addressing these issues may require not just architectural
 405 changes but also strategies like data reweighting or augmentation.

6 Discussion

Our analysis reveals multiple conceptual blindspots in four popular generative image models. The results presented here, however, only scratch the surface: each individual finding could warrant its own dedicated investigation. Rather than delving deeply into any one of these questions, we instead showcase the versatility of our method and exploratory tool. Out of the box, they allow for a systematic identification of concepts that models struggle to generate, detection of memorization artifacts, discovery of datapoints with insufficient captions, quantification of post-training effects, and characterization of conceptual shifts across model architectures. We thus open space for follow-up work to extend the depth of analysis, scope of evaluated architectures, and inquiry into root causes of conceptual blindspots. Future work could also explore hierarchical representations of concepts in RA-SAE to allow for a more nuanced analysis.

Limitations. We wish to highlight several limitations of our work. By relying on DINOv2 and RA-SAE for concept extraction and representation, our approach is inherently constrained to the kinds of concepts these models capture; concepts poorly represented by them will escape our analysis. Additionally, while our sample size of 10,000 images is substantial, it may not fully capture the long tail of rare concepts, concept co-occurrence, or other compositional statistics.

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Appendix

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A Experimental Setup

This section details the experimental setup for our analysis of four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Due to concerns with CSAM and other unsafe content in the dataset, the original data release is no longer available. A substitute release of a subset of this dataset with additional filtering of the unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe> is used.

The sampling procedure consists of: (1) loading the full LAION dataset using the Hugging Face `datasets` library, (2) performing validation to ensure proper URL structure and resource availability via HTTP HEAD requests, and (3) employing random sampling with replacement until reaching the target count of 10,000 valid samples. This approach, yielding D_G with (x, t) tuples, ensures our observation space contains accessible image-text pairs for comparative analysis of a dataset of image URLs whose large portion has been made unavailable since original release. Additional examples of synthesized images are shown in Appendix H.

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} to have a one-to-one correspondence with D_G , yielding triplets (x, x', t) . Specifically, given the 10,000 image-text pairs (x, t) from D_G , we use t to synthesize counterpart images x' using each generative model g_θ .

The synthesis process follows the standard text-to-image generation pipeline for each model architecture, implemented using the Hugging Face `diffusers` library, where the models are loaded at mixed precision (fp16). All synthetic images are generated at 512×512 pixel resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>.

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1. We compute energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated model. The sigmoid transformation with temperature $T = 0.8$ is applied during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 We embed the complete set of 32,000 concepts into two-dimensional space using UMAP applied to the sparse concept codes. Each point in this UMAP represents an individual concept, colored according to its energy difference $\delta(\cdot)$, emphasizing both suppressed and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlation coefficients between $\delta(\cdot)$ vectors of all model pairs, producing both scatter plots and correlation matrices. This analysis reveals whether blindspots cluster in conceptual space and identifies model-specific versus universal patterns of conceptual blindspots.

Section 5.3 We rank all 32,000 concepts by their energy difference $\delta(\cdot)$, and manually examine the extrema (both suppressed and exaggerated blindspots). For suppressed blindspots, we select concepts with $\delta(\cdot) < 0.1$; for exaggerated blindspots, we choose those with $\delta(\cdot) < 0.9$. Presented examples are manually annotated with textual descriptions of the respective concepts through inspection of their most activating images and spatial attention patterns. We outline ongoing efforts to automate this concept interpretation in Appendix D.

Section 5.5 We compare 1.5 with and without DPO in the following fashion: for each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. We apply a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before taking their mean. This yields datapoint-wise energy differences that quantify how much each generated image deviates from its natural counterpart in concept space. Finally, these differences are visualized as overlapping histograms, contrasting both model variants.

Section 5.6 For each concept c_k , its empirical frequency $\|Z_{:,i}\|_0$ (the count of non-zero activations across the natural dataset) is counted. A sigmoid normalization with temperature $T = 0.4$ is then applied to the energy differences $\delta(\cdot)$. The analysis is visualized using scatter plots where the x-axis is the empirical concept frequency and the y-axis is the sigmoid-transformed energy difference. The point sizes are proportional to activation frequency and point colors are proportional to the magnitude of energy differences.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. This yields a scalar measure of conceptual divergence for each image pair. The samples are ranked by their energy differences. Minimal divergence indicate potential memorization artifacts and maximal divergence point to significant conceptual failures. This analysis enables qualitative inspection of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized data D_G and D_{go} , concept co-occurrence patterns are analyzed through the co-activation matrix $Z^T Z$, which holds pair-wise correlations in concept usage. Spectral analysis is performed to examine the dominant conceptual directions using eigendecomposition. The alignment between natural and synthetic co-occurrence structures is assessed using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence matrix. These 100×100 similarity matrices are visualized as square heatmaps where perfect diagonal alignment would indicate identical principal concept axes, while off-diagonal patterns reveal would revolve rotations and mismatches in compositional geometry.

935 B Custom RA-SAE Training Details

936 This section summarizes the training details of our custom archetypal SAE (RA-SAE) [56]. The
937 model is open-sourced at a software repository.

938 **Dataset.** The auto-encoder is trained on the complete ImageNet-1k training split, (≈ 1.28 M)
939 RGB images. Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are fed
940 to the SAE without class or position embeddings. The total number of training tokens is therefore
941 $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

942 **Dictionary.** The dictionary has 32,000 concept dimensions. For the sparse activation rule, top- k
943 masking with $k = 5$, is used; activations outside the largest five per input are set to 0. The weights
944 are initialized using Xavier/Glorot. The training is conducted at mixed precision (fp16), with the last
945 ten epochs performed at full precision.

946 **Optimizer and Schedule.** The model is trained for 50 epochs using base AdamW ($\beta_1 = 0.9$,
947 $\beta_2 = 0.999$) optimizer is employed with weight decay set to 10^{-5} . Linear warm-up is applied on the
948 first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. MSE loss is used
949 alongside an auxiliary term penalizing activations that never enter the top- k set, where $\lambda = 10^{-5}$.

C Computational Resources

This section summarizes the GPU resources used for training and experiments in support of this paper. In total, we used approximately 202 GPU-hours on NVIDIA H100s and H200s.

RA-SAE. Trained for approximately 24 GPU-hours on three NVIDIA H100s.

Synthesized Images. Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately 100 GPU-hours on a NVIDIA H200.

● **Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

● **Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one H200 GPU (total ~ 15 GPU-hours).

962 D Autointerpretability

963 In an effort to automate the interpretation of concepts identified by the RA-SAE, we qualitatively
 964 evaluated the ability of Vision-Language Models (VLMs) to describe them. In particular, we queried
 965 ChatGPT 4o via the OpenAI API using various prompts and formats of an image x'' in which the
 966 given concept was salient, recording the description generated by the VLM. In this section, we
 967 highlight the findings from our exploration. For evaluation purposes, we allowed the VLM to provide
 968 its full reasoning; to make this a scalable solution, a simple modification of the prompt—asking the
 969 VLM to provide only the description—would filter out the reasoning.

970 First, we provided the VLM with two images: a raw version of x'' and x'' with a heatmap superim-
 971 posed to localize the intensity of the concept:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

972 As seen in the example, the VLM did not interpret the heatmap visualization correctly. Instead of
 973 describing the wood texture in the background—the actual salient region—it described the ring in the
 974 foreground, which received no attention from the RA-SAE.

975 To determine whether this was a prompting issue, we tested the same scenario again—this time using
 976 an expanded prompt that explained the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

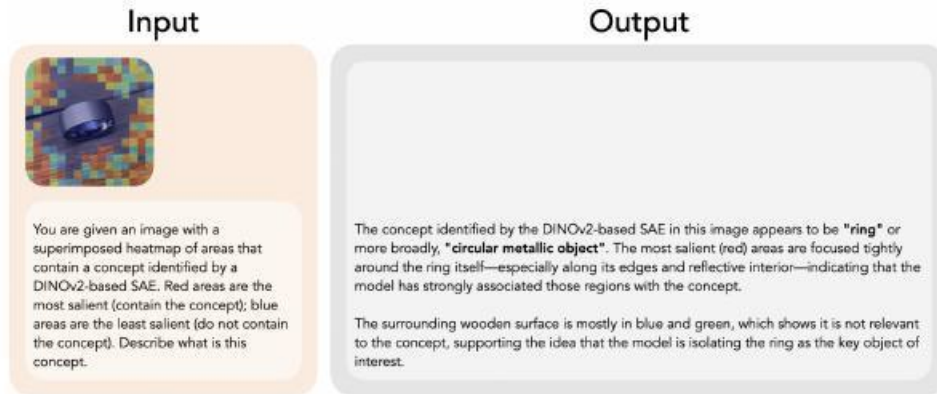
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

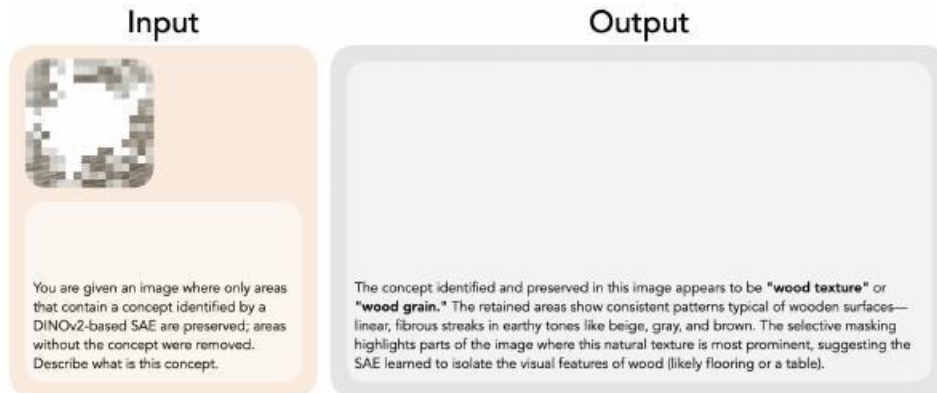
977 This prompt modification did not resolve the issue, as the VLM still identified the ring as the
 978 prominent concept in the image. Several other prompting techniques, including those describing the
 979 heatmap in even greater depth (e.g., by including the code used for its generation), were explored
 980 with similar results.

981 We next tested providing the VLM only with the heatmap-superimposed x'' and a similar prompt
 982 explaining how to interpret the heatmap:



983 Once again, the result failed to identify the wooden background as the prominent concept in the image
 984 and instead focused on the ring. As in the previous case, we experimented with various prompting
 985 strategies using this image format, but the results remained largely unchanged.

986 From these observations, we concluded that the heatmap alone is insufficient to guide the VLM's
 987 attention to the salient areas. The presence of less relevant regions appears to distract the model. To
 988 address this, we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :



989 As shown in the example, the VLM was now able to correctly identify wood texture as the dominant
 990 concept. We observed consistent success across a range of concepts and images.

991 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
 992 evaluation. However, we note that this evaluation is limited by its qualitative nature (due to the
 993 absence of ground truth annotations) and its focus on a single VLM. We hope future work on the
 994 autointerpretability of SAE concepts can build on and expand this analysis.

995 E Additional Results: Model-Specific Blindspots

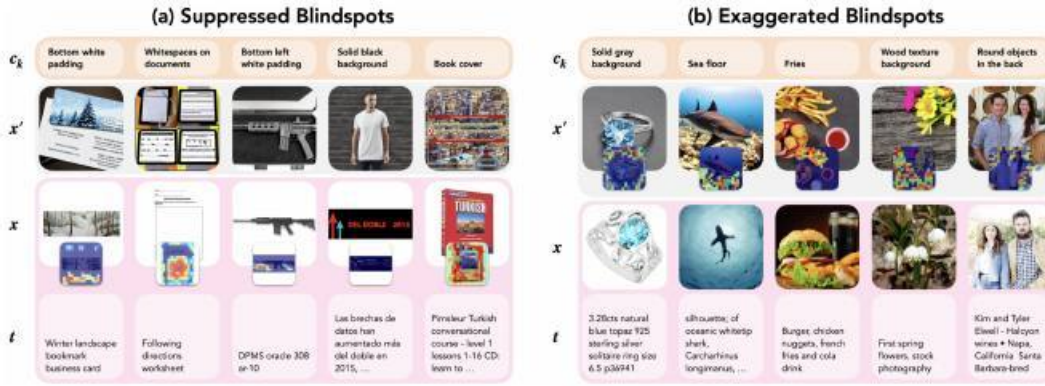


Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

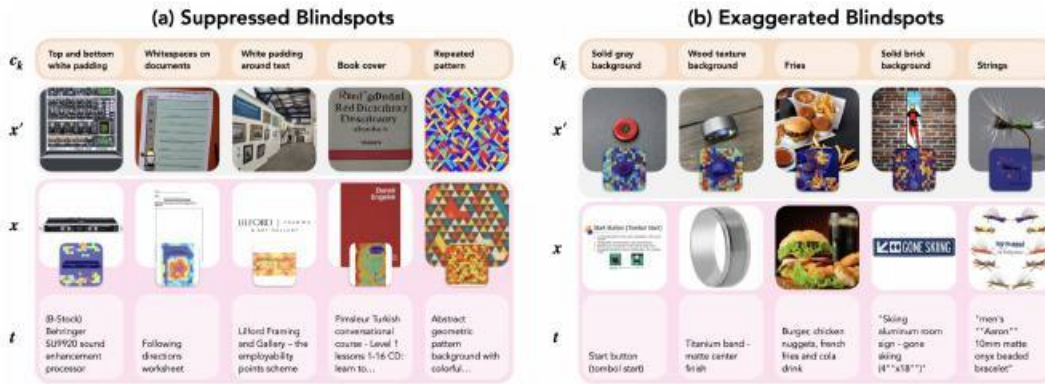


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

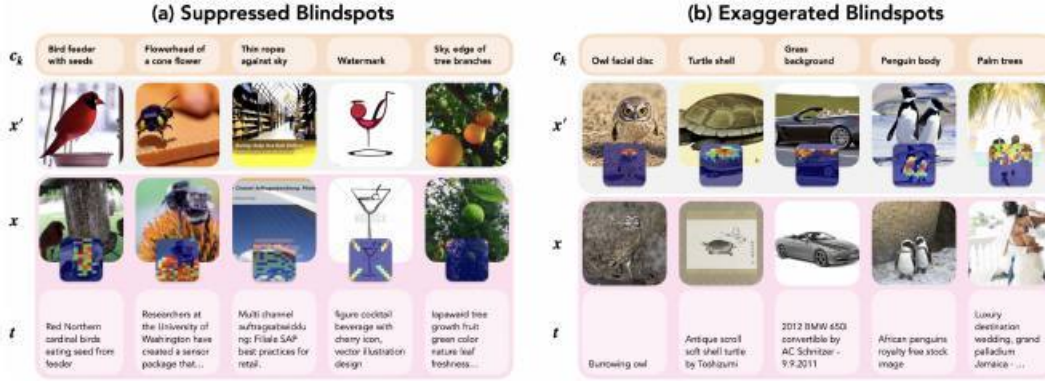


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, our analysis has centered on individual concept activations. Yet visual scenes are rarely composed of isolated concepts; instead, they are structured through rich and structured co-occurrence patterns that encode compositional semantics. We now examine whether generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which reflects pairwise correlations in concept usage.

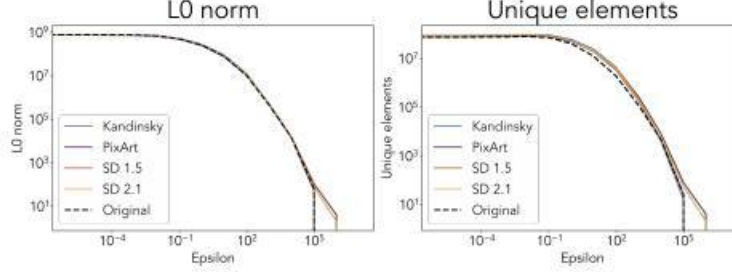


Figure 18: **Sparsity and Structural Divergence.** On the left: L0 norm of the co-occurrence matrix $Z Z^T$ as a function of ϵ (threshold), indicating how many entries remain active in each model. On the right: Number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure, but diverge in activation content.

Surprisingly, when assessed at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the ℓ_0 norm of $Z^T Z$ – thresholded at varying ϵ values – tracks closely between the natural and synthesized distributions across all models. This indicates that the gross connectivity of the conceptual graph, i.e., which concepts tend to co-activate at all, is well preserved. Formally, one can deem $Z^T Z$ as the adjacency matrix of a weighted, undirected graph over concepts, where edge weights reflect co-activation strength across the dataset.

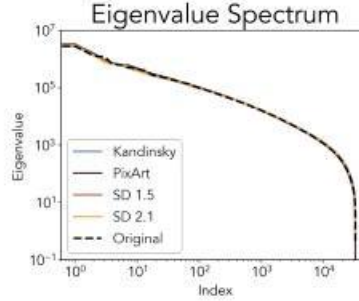


Figure 19: **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra from co-occurrence matrices $Z Z^T$ across models and the natural distribution. All evaluated models match the heavy-tailed decay of the natural distribution.

However, as illustrated in Fig. 18 (right), the specific content of these co-activations diverges: a substantial portion of entries in the model-generated $Z^T Z$ are not shared with the natural baseline. This suggests that while the capacity for compositionality is retained, the identity of active pairings may shift, potentially reflecting model specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 shows the eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. All spectra exhibit a heavy-tailed decay, consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

Further, we examine the alignment of dominant conceptual directions via cosine similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). While all models exhibit partial diagonal alignment—implying overlap in principal concept axes—the

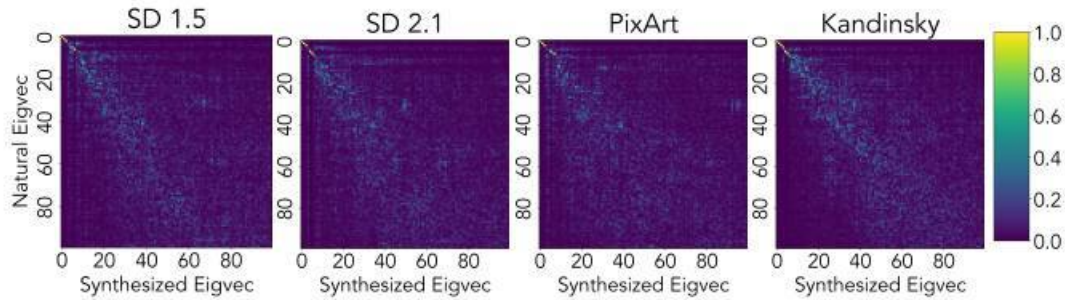


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure shows alignment of dominant conceptual directors, with varying degrees of alignment across the four models.

1022 off-diagonal entries reveal rotations and mismatches in higher modes, reflecting evident deviations in
 1023 compositional geometry.

1024 Together, these findings reveal that diffusion models approximate the global shape of concept co-
 1025 activation surprisingly well, yet deviate in subtle and structured ways when examined through
 1026 the spectral lens. Such higher-order discrepancies may underpin failures in generating coherent,
 1027 multi-object scenes or relational concepts.

1028 G Additional Results: Blindspot Stress Testing

1029 To stress-test the blindspots identified by our method, we gathered a range of prompts describing
 1030 these blindspots and used them to generate many images. We then contrasted the outputs from models
 1031 in which the concept was identified as a blindspot with those in which it was not.

1032 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1033 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1034 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per
 1035 prompt and analyzed using our custom RA-SAE model (see Appendix B), which ranked them by
 1036 the intensity with which the desired concept appeared. All images were then manually reviewed to
 1037 determine whether the blindspot was successfully depicted.

1038 As seen in the following examples of suppressed concepts, while some aspects of the target concept
 1039 occasionally appeared (e.g., a holder or string for the bird feeder blindspot and a round hole for
 1040 the glossy DVD disc blindspot), the models generally failed to generate the full concept. This
 1041 aligns with our method’s assessment and supports the validity of the stress test.

1042 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Examples of images generated with various prompts involving the bird feeder concept as a part of the stress testing. In Kandinsky, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, SD 1.5, in which this concept was not identified as a blindspot, is able to generate this concept.



Figure 22: Examples of images generated with various prompts involving the glossy DVD disc concept as a part of the stress testing. In SD 1.5, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, Kandinsky, in which this concept was not identified as a blindspot, is able to generate this concept.

1044 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pair examples from LAION-5B with matching images generated with the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1045 I Concentration bounds for δ

1046 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1047 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To
 1048 address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid's
 1049 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,
 1050 we can obtain fast and accurate estimates of concept bias.

1051 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1052 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1053 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1054 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1055 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1056 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1057 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1058 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1059 By McDiarmid's inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1060 which proves the claim. □

1061 Practically, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1062 a few reaching values up to 10. The concentration bound shows that even for the largest observed
 1063 biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that deviate from the true value by
 1064 no more than a small ε with high probability. This justifies our sampling strategy and confirms that
 1065 accurate bias measurements are attainable with limited data.

1066 J Monotonicity and Calibration-Free Interpretation of δ

1067 Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias. In
 1068 practice, we use the score $\delta(k)$ for this purpose. However, one may wonder whether such a score
 1069 introduces distortions relative to more direct quantities such as the energy gap or the odds ratio.
 1070 The following result establishes that $\delta(k)$ is a strictly increasing reparameterization of both, and
 1071 therefore inherits their ordering. This guarantees that no calibration is needed when using $\delta(k)$ to
 1072 rank concepts.

1073 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1074 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1075 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1076 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1077 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1078 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1079 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1080 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1081 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1082 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1083 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1084 concepts.

1085

□

1086 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual
 1087 generation odds ρ_k . No calibration is necessary, and all three quantities preserve the same total
 1088 ordering over concepts.

1089 K Stability of FID Under SAE Embeddings

1090 In this section we establish a quantitative relationship between the Fréchet Inception Distance
 1091 (FID) computed in the original activation space (of dimension d) and the FID after applying a
 1092 (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D
 1093 has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1094 We start by recalling a simple fact: if D is not overcomplete, orthogonal and $k = d$, then we have an
 1095 isometry between A and Z , implying that the FID is perfectly preserved. However, this case is not
 1096 realistic, we will then turn to the overcomplete case, and show we can bound FID by the extreme
 1097 singular value of D . We will work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is just $\mathcal{W}_2^2$
 1098 specialised to Gaussians.

1099 For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) =$
 1100 $\mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently
 1101 the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1102 Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof does not rely on that. We will start by a
 1103 simple lemma in the case where D is not overcomplete.

1104 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
 1105 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
 1106 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1107 *Proof.* Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes
 1108 with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$
 1109

1110 This case, however, is quite unrealistic as SAE usually rely on the overcompleteness to extract
 1111 meaningful and interpretable concepts. In the overcomplete case, D is no longer orthonormal, but
 1112 we can still have column-orthonormal dictionary. We will use that to show that we can bound using
 1113 the extremal singular value of $D^\top D$.

1114 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
 1115 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
 1116 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1117 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1118 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
 1119 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1120 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1121 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
 1122 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1123 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1124 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1125 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1126 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1127 finishes the proof. $\square$

1128 Essentially, theorem [6](#) tells us that applying a column-orthogonal overcomplete SAE dictionary
 1129 cannot distort Fréchet Inception Distance by more than the square of its extremal singular values.
 1130 When $\mathbf{D}$ is nearly orthogonal – empirically we usually found that $\sigma_{\min}, \sigma_{\max} \approx 1$ – the result implies
 1131 that FID measured in the SAE feature space is essentially close to the canonical FID. $\square$

1132